



THE UNIVERSITY
OF QUEENSLAND
AUSTRALIA

CREATE CHANGE

Secure AI Workflow for Confidential Research

Authorized evidence, controlled tools, and accountable collaboration

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Research Problem

How can researchers use AI and scientific software without exposing confidential data?

Keep data inside

Files and model context stay inside the protected network.

Use permitted evidence

Only approved evidence reaches the model.

Control every action

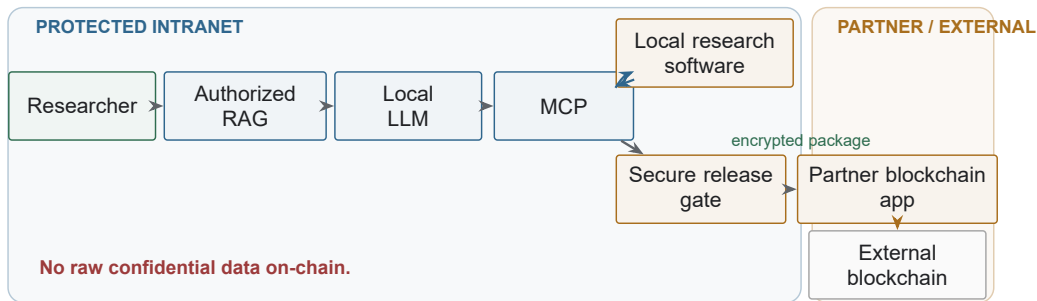
Narrow interfaces control tools and release.

One researcher experience; several independent trust boundaries.

Target Workflow

MCP is the controlled bridge between the local AI, research software, and external collaboration.

TARGET DESIGN

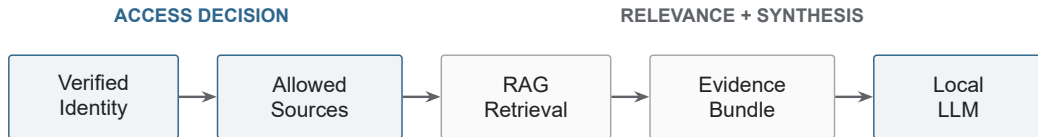




Authorize Before Retrieval

Identity and policy grant access; RAG only ranks evidence inside that approved scope.

TARGET DESIGN



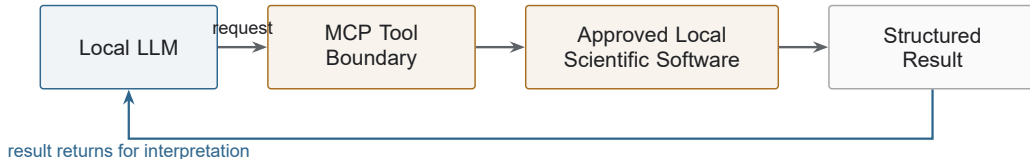
The model never decides access.

Failed authorization stops retrieval.

MCP as the Tool Boundary

The model proposes a bounded tool call; approved local software performs the scientific operation.

PROPOSED; NOT IMPLEMENTED



Approved tools

Structured I/O

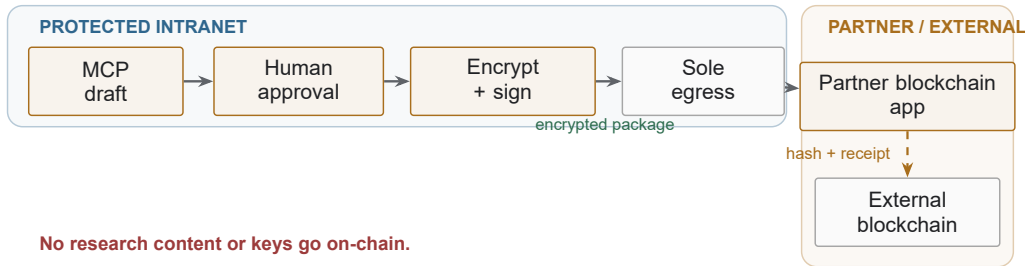
Human review



Confidential Collaboration

Confidential payloads move off-chain as protected packages; the ledger stores only a minimal proof.

CONCEPT; INTERFACE UNDEFINED

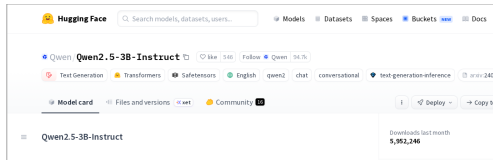


No research content or keys go on-chain.

Model Options

Model scale changes deployment; it does not replace authorization or review.

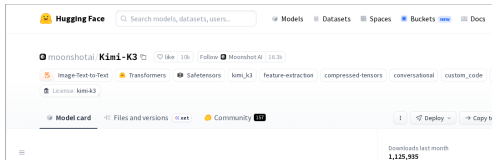
LOCAL-SCALE CANDIDATE



Qwen2.5-3B-Instruct

3B smoke-tested on one synthetic photonics fixture;
not scientific validation.

CLUSTER-SCALE CANDIDATE



Kimi K3

Open-weight; 2.8T / 104B active; cluster-only.

CURRENT EVIDENCE

**Current smoke test: qwen2.5:3b via loopback Ollama; qwen2.5:7b remains the handoff baseline.
Neither candidate is enterprise integrated.**



What Runs Today

Only the local evidence workflow is operational; the intranet, MCP, and blockchain paths remain future work.

RUNS NOW

Files → traceable evidence → local LLM → checked report

Single user, local access; public or synthetic data.

UNDER REVIEW

Stable interface between the agent and local model

NOT BUILT

Authorized RAG | MCP tools
Secure release | Partner ledger

Engineering Validation Path

Validate each trust boundary before connecting scientific tools or external networks.



IMMEDIATE CHECKPOINT

Complete the local model-interface review.